

The Minimal Polynomial, Cyclic Subspaces, and Jordan Normal Form

Math 5250 — Week 4

Polynomials over a Field

- ▶ Over an algebraically closed field, $f(x) = \prod_i (x - \lambda_i)^{n_i}$, where the λ_i are the roots and n_i is the multiplicity of λ_i .
- ▶ If $f(x) = P_A(x)$ and all $n_i = 1$, then A is diagonalizable.
- ▶ $F[x]$: polynomials with coefficients in F (not necessarily algebraically closed). $f(x) = a_0 + a_1x + \cdots + a_nx^n$ has degree n if $a_n \neq 0$, and is **monic** if $a_n = 1$.

The Division Algorithm

Proposition (Division Algorithm). For $f, g \in F[x]$, there exist unique polynomials $q(x), r(x)$, with $0 \leq \deg(r(x)) < \deg(g(x))$, such that

$$f(x) = q(x)g(x) + r(x).$$

Definition. $u(x)$ is a **divisor** of $f(x)$ if $f(x) = u(x)v(x)$ for some $v(x)$. $h(x)$ is a **common divisor** of $f(x)$ and $g(x)$ if it divides both.

Euclid's Algorithm

Proposition (Euclid's Algorithm). Given nonzero $f(x), g(x) \in F[x]$, there is a unique nonzero monic polynomial $d(x) = (f(x), g(x))$ of minimal degree of the form $a(x)f(x) + b(x)g(x)$. This **greatest common divisor** satisfies:

1. $d(x)$ is a common divisor of $f(x)$ and $g(x)$;
2. if $h(x)$ is a common divisor of $f(x)$ and $g(x)$, then $h(x)$ divides $d(x)$.

Roots of the GCD

Proposition. The roots of $d(x) = (f(x), g(x))$ are exactly the common roots of $f(x)$ and $g(x)$. If λ is a common root of multiplicity a in f and b in g , its multiplicity in $d(x)$ is $\min(a, b)$.

Corollary. $f(x)$ and $g(x)$ have no common root in an algebraic closure of F if and only if $a(x)f(x) + b(x)g(x) = 1$ for some polynomials $a(x), b(x)$.

Irreducible Polynomials

Definition. $f(x)$ is **irreducible** if it cannot be factored as $f(x) = g(x)h(x)$ with $\deg(g), \deg(h) \geq 1$.

Proposition. If $f(x)$ is irreducible and $a(x)b(x)$ is a multiple of $f(x)$, then $a(x)$ is a multiple of $f(x)$ or $b(x)$ is.

Unique Factorization

Proposition. A monic polynomial $f(x)$ factors as

$$f(x) = \prod_i p_i(x)^{n_i}$$

where the $p_i(x)$ are distinct monic irreducible polynomials, and the set of p_i and the n_i are unique up to order.

Decomposition

Let $L : V \rightarrow V$ be linear and factor $P_L(x)$ into powers of distinct irreducibles:

$$P_L(x) = p_1(x)^{n_1} \cdots p_k(x)^{n_k}.$$

Proposition. V decomposes as

$$V = V_1 \oplus V_2 \oplus \cdots \oplus V_k$$

where $p_i(L)^{n_i} V_i = 0$ for each i .

The Minimal Polynomial

Let V be finite dimensional and $L : V \rightarrow V$ linear. By Cayley-Hamilton, $P_L(L)V = 0$.

Definition. The **minimal polynomial** $m_L(x)$ is the monic polynomial of smallest degree such that $m_L(L)V = 0$.

Proposition. $m_L(x)$ is a divisor of the characteristic polynomial $P_L(x)$.

Cyclic Subspaces

Definition. A finite dimensional vector space V of dimension n is a **cyclic space** for $L : V \rightarrow V$ if there is $v \in V$ such that $L^i(v)$, $i = 0, \dots, n-1$, span V . Such a v is a **generator** for V .

If v is a generator, there is a relation $L^n v = -\sum_{i=0}^{n-1} a_i L^i v$, giving a polynomial

$$q(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$$

with $q(L)V = 0$; $q(x)$ is the minimal polynomial of L on V .

The Companion Matrix

In the basis $\{L^i v\}_{i=0}^{n-1}$, the matrix of L is

$$A = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & 0 & \cdots & 0 & -a_2 \\ 0 & 0 & 1 & \cdots & 0 & -a_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix}$$

Remark. A matrix of this shape for $q(x)$ is called the **companion matrix** of $q(x)$.

Proposition. The characteristic polynomial of A is $q(x)$.

Corollary. If V is cyclic for L , then the minimal and characteristic polynomials of L coincide.

Decomposition into Cyclic Subspaces

Theorem. Let $L : V \rightarrow V$ be linear, where V has dimension n over F . Then there is a decomposition

$$V = V_1 \oplus V_2 \oplus \cdots \oplus V_k$$

into cyclic subspaces, where the minimal polynomial of V_i is a power $p_i(x)^{k_i}$ of a monic irreducible polynomial over F . The $p_i(x)^{k_i}$ are unique up to order.

Consequences of the Decomposition

- ▶ The minimal polynomial $m_L(x)$ of V is the least common multiple of the $p_i(x)^{k_i}$.
- ▶ The characteristic polynomial of V is the product of the characteristic polynomials of the V_i , i.e. the product of the $p_i(x)^{k_i}$.
- ▶ Every irreducible factor of $P_L(x)$ is a factor of the minimal polynomial of L .

The Jordan Normal Form

Suppose $P_L(x) = \prod_i (x - \lambda_i)^{n_i}$ (always true if F is algebraically closed, but not required). The cyclic decomposition writes V as a sum of cyclic subspaces V_i with minimal polynomial $(x - \lambda_i)^{k_i}$.

Definition. The $k \times k$ matrix with λ on the diagonal, 1 on the subdiagonal, and 0 elsewhere is the **Jordan block** $J_k(\lambda)$:

$$J_k(\lambda) = \begin{pmatrix} \lambda & 0 & 0 & \cdots & 0 & 0 \\ 1 & \lambda & 0 & \cdots & 0 & 0 \\ 0 & 1 & \lambda & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & \lambda \end{pmatrix}$$

The Jordan Normal Form (cont.)

Theorem (Jordan Normal Form). If the characteristic polynomial of L factors into linear factors, there is a basis of V in which the matrix of L is block diagonal, with blocks $J_{k_i}(\lambda_i)$, where the $(x - \lambda_i)^{k_i}$ are the minimal polynomials of the cyclic subspaces V_i .

Remark. Equivalently, $A = D + N$ where D is diagonal with the eigenvalues of A (repeated by multiplicity) and N is nilpotent ($N^k = 0$ for some k). In a single Jordan block of dimension k , $A = \lambda I + N$ with $N^k = 0$ but $N^{k-1} \neq 0$.

Remark. Each Jordan block has a one-dimensional space of eigenvectors, so the number of independent eigenvectors equals the number of Jordan blocks.